NAO VI Guide

Explanation of Flowchart

Last Modified Date: 10/12/2025

For:

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Introduction

This document provides an explanation of the flowchart illustrated in the accompanying *Flowchart* PDF. The purpose of this document is to describe the sequence of actions, decision points, and logical flow used in the NAO robot's interactive behavior system. Each section of the chart—covering startup, autonomous routines, tactile and voice command inputs, and response actions—is detailed to ensure a clear understanding of how the system transitions between different states during operation. This explanation aims to serve as a reference for development, debugging, and demonstration within the NEST Lab at the University of British Columbia.

Flowchart

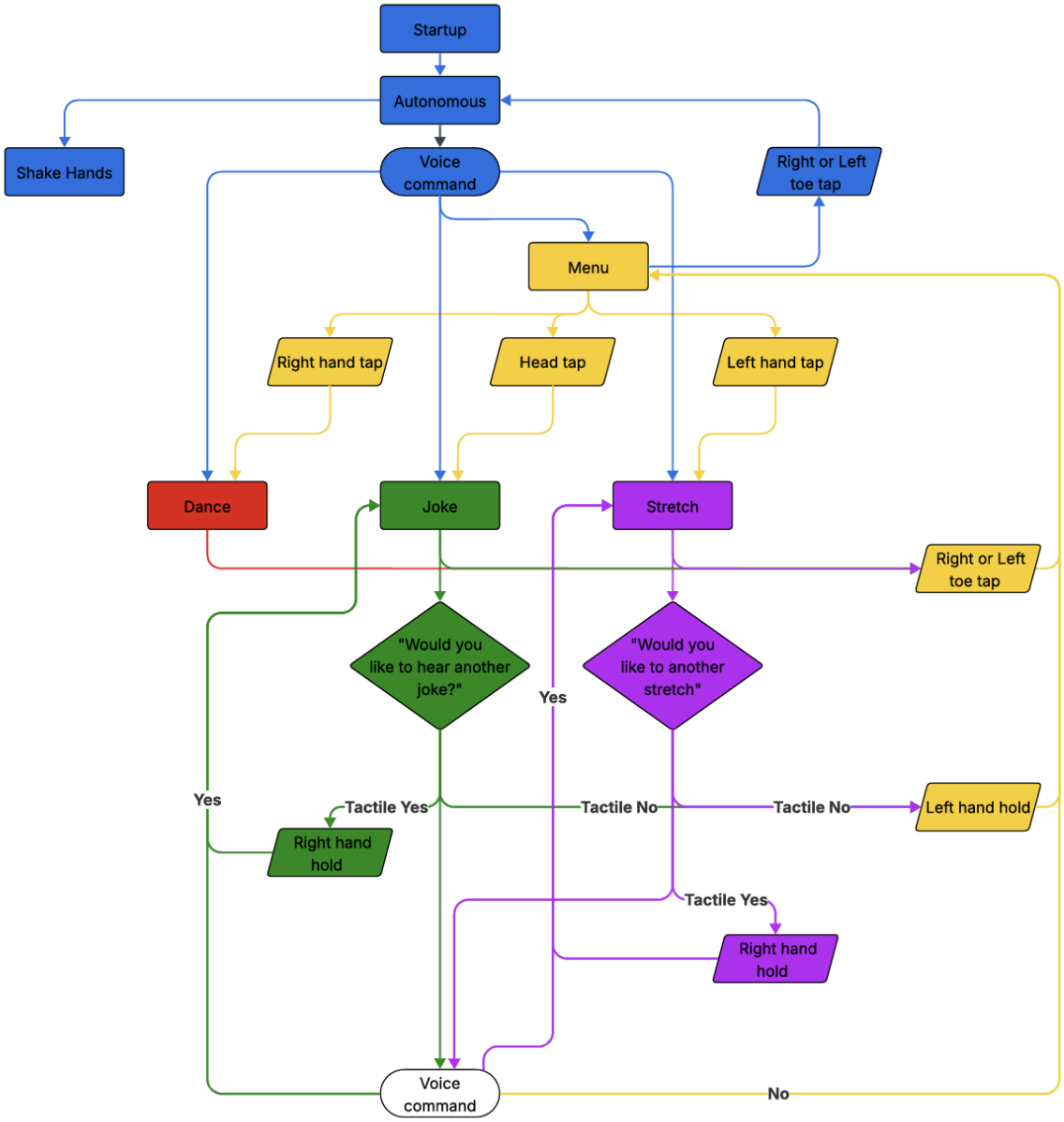

Figure 1 Flowchart for NAO behaviors

Instructions

Blue Section – Initialization and Core Control

🟦 The process begins at the **Startup** stage, where the NAO robot powers on and initializes all systems. It then enters the **Autonomous** mode, allowing it to independently recognize and respond to both tactile and voice inputs. In this mode, the robot can directly take **voice commands** to perform actions such as 🟥 **Dance**, 🟩 **Joke**, or 🟪 **Stretch** without requiring manual input.
If the user says the keyword **"Menu,"** the robot transitions into a selection interface where tactile gestures are used to choose activities: a **Right Hand tap** activates **Dance**, a **Head tap** triggers **Joke**, and a **Left Hand tap** starts the **Stretch** routine. This design enables both hands-free operation through speech and manual selection via tactile interaction, providing flexible user control.

Yellow Section – Menu and Gesture Inputs

🟨 The **Menu** section acts as a central hub for physical triggers. It can be reached either from a voice command or a toe tap. Within this area, the robot reacts to three tactile cues: **Right Hand tap**, **Head tap**, and **Left Hand tap**. Each gesture leads to a different action block: the **Right Hand tap** initiates a **Dance**, the **Head tap** begins a **Joke**, and the **Left Hand tap** starts a **Stretch** routine. This section defines how NAO interprets human touches to select among different modes of interaction.

Red Section – Dance Routine

🟥 The **Dance** block, activated by **voice command** or **Right Hand tap**, represents a simple performance behavior. Once the dance is completed, the flow returns to the **Voice command** or **Menu** pathway, allowing the user to initiate another interaction. A **force stop** and **return to menu** can be performed by **tapping left or right toe** during the dance.

Green Section – Joke Routine

🟩 When the user triggers the **Joke** routine (by **voice** or **Head tap**), NAO tells a joke and then asks, *"Would you like to hear another joke?"* If the tactile response is **Yes** (through **voice** or a **Right Hand hold**), the robot loops back to the joke stage to tell another one. If the tactile response is **No** (**Voice** or **Left Hand Hold**), control returns to the **Voice command** stage for a

new instruction. This green segment captures the humor and dialogue interaction pathway. a **Right or Left toe tap** at this stage can also redirect the system to the **Menu**.

Purple Section – Stretch Routine

🟪 The **Stretch** routine begins either from a **Left Hand tap** or from **Voice**. After completing one stretch cycle, NAO asks, *"Would you like to do another stretch?"* A **Yes** response (**Voice** or **Right Hand hold**) restarts the stretching motion, while a **No** (**Voice** or **Left Hand hold**) response transitions control back to **Voice command**. Additionally, a **Right or Left toe tap** at this stage can also redirect the system to the **Menu**, allowing users to easily switch tasks.